



Fig. 6: Key generated for posting data on ThingSpeak



Fig. 7: Data posted on ThingSpeak API

com/products/mcu/Pages/USBtoUARTBridgeVCPDrivers.aspx)

- Smart Wi-Fi comes with the latest version of NodeMCU firmware pre-loaded
- NodeMCU API (https://github.com/nodemcu/nodemcu-firmware/wiki/nodemcu_api_en)

We describe two tests (TEST1, TEST2) in this article. Before downloading the code in Smart Wi-Fi, you will need to connect it to the PC using a micro-USB cable. Start ESPLorer tool and set the port and baud rate [9600 default] and press Open Tab to open the port. It is ready to use now. We have used ESPLorer v0.2.0-rc2 version, and the corresponding test procedure is described below.

Test 1 (Temperature sensor interface with Smart Wi-Fi)

The code here is for data read on the ADC and sent to the serial port locally. Open `Temperature_Interface.lua` file (given in DVD), type `= call("Temperature_Interface.lua")` on the left side of Send and press Send.

We can now observe data into the serial terminal of ESPLorer tool. The code will keep sending data over the serial port every 15 seconds. In the

code, we use internal timer # 1 of ESP to call an inline function every 15 seconds. The function reads ADC0 value, which it receives in terms of millivolts, and divides it by ten. As the resolution of LM35 is 10mV per degree Celsius, dividing by ten provides the actual temperature in Celsius. We can print this value to the serial port using print function:

```
tmr.alarm(1, 15000, 1, function()
current_temp = adc.read(0)/10;
print(fmt("Temperature : %3.1f C", current_temp));
end);
```

Test 2 (Temperature data upload on the Internet using ThingSpeak API)

In Test 1, we connected a temperature sensor and observed the data locally. This data can be used in multiple ways as per the application. In Test 2, we will make the same data available on the Internet and it would be accessible from anywhere in the world.

We use ThingSpeak API, which is a freely-available tool used to access and present data over the Internet. Before using ThingSpeak API, you need to create a user account for ThingSpeak and a channel. You will get an API key (Fig. 6) for the channel and then you can upload your data and present it in a graphical form. The only limitation is the frequency of posts, which is limited to 15 seconds.

Before going ahead with data upload, the local Wi-Fi router needs to be paired with Smart Wi-Fi. Use the following command to connect it to your Wi-Fi router:

```
call("connectap","your router SSID","your router password");
```

This command will pair Smart Wi-Fi to the Wi-Fi router you want to connect with. The response will be the IP address assigned to Smart Wi-Fi by the router.

Open `ThingSpeak_upload.lua` file (given in DVD), type `= call("ThingSpeak_upload.lua")`

on the left side of Send and press Send. Data output on ThingSpeak API is shown as:

Error: Reference source not found.

The application code that will post data to ThingSpeak server (Fig. 7) is given below.

```
myKey = "put your API key here"
function post()
call("setLED", 2);
field1 = adc.read(0)/10;
print(fmt("%3.1f",field1));
call("httpsend","api.thingSpeak.com", 80, "update?api_key=..myKey..&field1=..field1");
call("setLED", 0);
tmr.delay(50);
end
call("connectap","your router SSID","your router password")
tmr.alarm(1, 15000, 1, post)
```

Applications of Smart Wi-Fi

Smart Wi-Fi is an IoT-enabler tool. The applications it can cater to are only limited by the imagination of makers. The very basic applications could be for smart homes or smart offices.

This module can be used for data logging, data monitoring and more, and provides very good support for product development. It also has all features to act as a full-fledged product.

Conclusion

Using Smart Wi-Fi module, we have seen how easy it is to interface and connect sensors to the Internet. Tools like ESPLorer and open source LUA scripts make development very easy and fast. Platforms such as ThingSpeak add to the benefits and provide support for testing and development of an IoT product. **RFY**



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